

WaiYan (Anson) Chan

Ph.D. Candidate / Graduate Research Assistant

University of Wisconsin-Madison

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EDUCATION

University of Wisconsin-Madison

Ph.D. in Electrical and Computer Engineering

Expected 2028

- Thesis: Design and Control for Over-Actuated Lightweight Electric Machines
- Minor: Mechanical Engineering
- Advisor: Lei Zhou

University of Wisconsin-Madison

M.S. in Electrical and Computer Engineering

Sep. 2022 – Dec. 2021

- Project: Scaling of High-Speed Surface PM Bearingless Machines
- Advisor: Eric L. Severson

University of Wisconsin-Madison

B.S. in Mechanical Engineering

Sep. 2017 – Dec. 2021

- Project: Compact Low Voltage Induction Machine with 3D Printed Housing
- **Dean's Honor List**

RESEARCH

Precision Mechatronics and Control Lab

EXPERIENCE

Graduate Research Assistant

Dates

Severson Research Group

Graduate Research Assistant

Dates

Project:

- Additively Manufactured (AM) Stator Housing for High-Speed Bearingless Generator
 - Designed and fabricated AlSi10Mg AM motor housing with integrated cooling channels for a 100kW, 80kRPM twin bearingless generator for a combined heat and power (CHP) generation system.
- Bearingless Machines (BSPM) for Aerial E-Turbocharger Application
 - Conducted structural and modal analysis to validate rotor design by identifying critical speed and rotor stress.
 - Modeled and conducted FE analysis of an 8kW 4-DOF twin-stator BSPM, along with component fabrication.
 - Facilitated the development of multi-physics modeling framework for BSPM and created Python scripts for evaluating machine constants and coil inductances in JMAG.

Wisconsin Electric Machine and Power Electronic Consortium

Undergraduate Research Assistant

Dates

- Prototyped a desktop size dynamometer setup capable of characterizing electric machines rated up to 10 N-m of torque and 3000 RPM. Reduced the original setup volume by 35%.
- Collaborated with a team of graduate researchers to prototype a modular terminal box for powering a 200 kW NASA prototype machine. Responsible for sourcing parts and creating CAD models.

- Fabricated BSPM machines by producing stator slot windings and machining components used in the assembly.

INDUSTRY EXPERIENCE	Motibera, Inc.	
	Motor Design Engineer Intern	May. - Aug. 2025
	Milwaukee Tool	
	NPD Mechanical Design Engineer	Jan. - Aug. 2022
- Conducted FE analysis on BLDC machines of various sizes and winding configurations; recommended optimal designs for product development using a Pugh Matrix to support informed and cost-effective decision-making. - Assessed performance requirements of competitor power tools by collecting and analyzing motor thermal characteristics and power output data.		
ADDITIONAL EXPERIENCE	Copmany Name	
	Title	Dates
- Description - Description - Description		
	Additional Title	Dates
	Additional Title	Dates

GRANTS AWARDED	University College Something or Other Grant	Dec 2014
PUBLICATIONS	T. Noguchi, W. Chan, N. Petersen, L. Rapp, and E. Severson, "Opportunities to enhance sCO₂ power cycle turbomachinery with bearingless motor/generators," Solar Compass, vol. 12, p. 100094, 2024. W. Chan, T. Noguchi and E. L. Severson, "Scaling of High-Speed Surface PM Bearingless Machines," 2024 IEEE Energy Conversion Congress and Exposition (ECCE), Phoenix, AZ, USA, 2024, pp. 5631-5638. T. Noguchi, N. Petersen, W. Chan, L. Rapp, E. Severson, "Bearingless Motor/Generator Applications in sCO₂ Power Cycles," The 8th International Supercritical CO₂ Power Cycles Symposium, San Antonio, TX, USA, 2024 T. S. Slininger, W. Chan, E. L. Severson and B. Jawdat, "An Overview on Passive Magnetic Bearings," 2021 IEEE International Electric Machines & Drives Conference (IEMDC), Hartford, CT, USA, 2021, pp. 1-8.	
PRESENTATIONS	<i>Title</i> , Department of Mathematics, University College. (December 2015)	
PROGRAMMING LANGUAGES	Proficient: MATLAB, Bash, L ^A T _E X, Maple, LilyPond Familiar: JavaScript, Google Apps Script, Python, Mathematica, C++	
CERTIFICATIONS	Title, Organization	Month 2015